QP CODE:25901871

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-BVOC (HONOURS) Regular Examination – April 2026

Second Semester

Skill Development Components (SDC)

MG2SDCSDS100 – COMPUTER SYSTEM DESIGN & ARCHITECTURE

Duration: 2 hours
Maximum Marks: 70

PART A – Very Short Answer Questions

(Answer any 10 questions – Each carries 2 marks)

1. Define instruction format.

Answer:
Instruction format is the arrangement of bits in an instruction that specifies the operation code (opcode), operand/address fields, and addressing mode.

- Meaning of instruction format – 1 mark
- Mention of opcode/operand fields – 1 mark

2. What is a status flag?

Answer:
A status flag is a bit in the flag register used to indicate the result or condition of an arithmetic or logical operation, such as zero, carry, sign, or overflow.

- Definition – 1 mark
- Example/condition mention – 1 mark

3. Define Flash memory.

Answer:
Flash memory is a non-volatile semiconductor memory that can be electrically erased and reprogrammed.

- Non-volatile memory – 1 mark
- Electrically erasable/reprogrammable – 1 mark

4. Define Input-Output Interface.

Answer:
An Input-Output interface is hardware that connects peripheral devices with the computer system and enables communication between CPU and I/O devices.

- Interface between CPU and peripherals – 1 mark
- Communication/control function – 1 mark

5. What happens during the fetch cycle?

Answer:
During the fetch cycle, the CPU retrieves the next instruction from memory using the Program Counter and stores it in the Instruction Register.

- Fetch instruction from memory – 1 mark
- PC/IR mention – 1 mark

6. What is general register organization?

Answer:
General register organization is the arrangement of multiple registers connected through a common bus and ALU to perform data processing operations efficiently.

- Arrangement of registers – 1 mark
- ALU/common bus/function – 1 mark

7. What is FIFO replacement policy?

Answer:
FIFO (First In First Out) replacement policy removes the oldest page/block from memory first when replacement is required.

- FIFO meaning – 1 mark
- Oldest page replaced – 1 mark

8. Which transmission is faster for short distance?

Answer:
Parallel transmission is faster for short distances because multiple bits are transmitted simultaneously.

- Parallel transmission identified – 1 mark
- Reason – 1 mark

9. Define Von Neumann Architecture.

Answer:
Von Neumann architecture is a computer architecture in which data and instructions are stored in the same memory and share the same bus system.

- Common memory for data/instructions – 1 mark
- Shared bus/system concept – 1 mark

10. What is meant by CPU operation?

Answer:
CPU operation refers to the execution of instructions through fetch, decode, and execute cycles.

- Execution of instructions – 1 mark
- Mention of fetch-decode-execute – 1 mark

11. What is a register?

Answer:
A register is a small high-speed storage location inside the CPU used to store temporary data and instructions.

- Small/high-speed storage – 1 mark
- Temporary data/instruction storage – 1 mark

12. Give two examples of hybrid devices.

Answer:

1. Modem
2. Fax machine
   (Other valid examples may be accepted.)

- 1 mark each example

PART B – Short Answer Questions

(Answer any 4 questions – Each carries 5 marks)

13. Explain I/O and system instructions.

Answer:

I/O Instructions

I/O instructions are used for communication between CPU and peripheral devices.

Examples:

- INPUT – reads data from input device
- OUTPUT – sends data to output device

Functions:

- Data transfer
- Device control
- Status checking

System Instructions

System instructions control processor operation and system functions.

Examples:

- HALT
- INTERRUPT
- ENABLE/DISABLE interrupt

Functions:

- Control execution
- Manage interrupts
- System control

Definition of I/O instructions – 1 mark

Examples/functions – 1.5 marks

Definition of system instructions – 1 mark

Examples/functions – 1.5 marks

14. Explain the advantages of hardwired control.

Answer:

Hardwired control uses fixed hardware circuits to generate control signals.

Advantages:

1. High speed operation
2. Faster instruction execution
3. Suitable for RISC processors
4. Reliable performance
5. No microprogram memory required

- Definition – 1 mark
- Any four advantages – 4 marks

15. Compare PROM, EPROM and EEPROM.

Feature	**PROM**	**EPROM**	**EEPROM**
Expansion	Programmable ROM	Erasable Programmable ROM	Electrically Erasable Programmable ROM
Erasing Method	Cannot erase	UV light	Electrical signal
Reprogramming	Only once	Multiple times	Multiple times
Speed	Fast	Moderate	Slower
Flexibility	Low	Medium	High

Marking Scheme

- Correct comparison points (any five) – 5 marks

16. Compare centralized and distributed arbitration.

Centralized Arbitration	Distributed Arbitration
Single controller decides bus access	No central controller
Simpler design	More complex
Lower cost	Higher cost
Single point failure possible	More reliable
Faster for small systems	Better for large systems

- Any five valid comparison points – 5 marks

17. Explain instruction pipelining with suitable example.

Answer:

Instruction pipelining is a technique in which multiple instructions are executed simultaneously by dividing instruction execution into stages.

Stages:

1. Fetch
2. Decode
3. Execute
4. Memory Access
5. Write Back

Example:

While one instruction is being executed, another is decoded and a third is fetched.

Clock Cycle	Instruction 1	Instruction 2	Instruction 3
1	Fetch		
2	Decode	Fetch	
3	Execute	Decode	Fetch

Advantages:

- Improves CPU throughput
- Increases performance

Definition – 1 mark

Pipeline stages – 2 marks

Example/diagram – 1 mark

Advantage – 1 mark

18. Compare paging and segmentation.

Paging	Segmentation
Fixed-size blocks	Variable-size blocks
Divides physical memory	Divides logical memory
Causes internal fragmentation	Causes external fragmentation
Address = page number + offset	Address = segment number + offset
Transparent to programmer	Visible to programmer

- Any five valid comparison points – 5 marks

PART C – Essay Questions

(Answer any 2 questions – Each carries 15 marks)

19. Evaluate the impact of CPI and clock frequency on CPU performance.

Answer:

CPU performance depends mainly on:

1. Clock frequency
2. CPI (Cycles Per Instruction)

CPU Execution Time

$$CPU\ Time = Instruction\ Count \times CPI \times Clock\ Cycle\ Time$$

OR

$$Performance \propto \frac{Clock\ Frequency}{CPI}$$

Clock Frequency

- Number of clock cycles per second
- Higher frequency → faster execution

CPI

- Average cycles needed per instruction
- Lower CPI → better performance

Impact:

- High clock frequency improves speed
- Low CPI reduces execution time
- Performance depends on balance of both

Example:

Processor A:

- 3 GHz, CPI = 2

Processor B:

- 2 GHz, CPI = 1

Processor B may perform similarly because of lower CPI.

Factors affecting CPI:

- Instruction set complexity
- Pipeline efficiency
- Cache performance

Conclusion:

Efficient CPU design aims for high clock frequency and low CPI to improve overall performance.

Marking Scheme (15 Marks)

- Definition of CPI – 2 marks
- Clock frequency explanation – 3 marks
- Formula – 2 marks
- Performance relationship – 3 marks
- Example – 2 marks
- Conclusion/factors – 3 marks

20. Explain superscalar architecture with block diagram and working.

Answer:

Definition

Superscalar architecture allows execution of multiple instructions simultaneously in a single clock cycle using multiple execution units.

Block Diagram Components

- Instruction Fetch Unit
- Decode Unit
- Dispatch Unit
- Multiple ALUs

- Register File
- Cache Memory

Working:

1. Multiple instructions fetched together
2. Instructions decoded simultaneously
3. Dispatch unit sends instructions to available execution units
4. Parallel execution occurs
5. Results written back

Features:

Advantages:

Disadvantages:

Marking Scheme

- Definition – 2 marks
- Block diagram – 4 marks
- Working – 5 marks
- Advantages/disadvantages – 4 marks

21. Explain address structure in cache memory for direct, associative and set-associative mapping.

Answer:

1. Direct Mapping

Each memory block maps to exactly one cache line.

Address Structure:

| Tag | Line | Word |

Features:

- Simple
- Fast
- More conflict misses

2. Associative Mapping

Any memory block can be placed anywhere in cache.

Address Structure:

| Tag | Word |

Features:

- Flexible
- Fewer conflicts
- Expensive hardware

3. Set-Associative Mapping

Cache divided into sets.

Address Structure:

| Tag | Set | Word |

Features:

- Combination of direct and associative mapping
- Better performance

Comparison:

- Direct → simple
- Associative → flexible
- Set-associative → balanced

Marking Scheme

- Direct mapping explanation + structure – 5 marks
- Associative mapping explanation + structure – 5 marks
- Set-associative explanation + structure – 5 marks

22. Discuss interrupt handling mechanism step-by-step with priority schemes.

Answer:

Interrupt

An interrupt is a signal that temporarily stops CPU execution to service an important event.

Steps in Interrupt Handling:

1. Device sends interrupt request
2. CPU completes current instruction
3. CPU saves program state
4. Interrupt acknowledged
5. ISR (Interrupt Service Routine) executed
6. Interrupt serviced
7. CPU restores previous state
8. Main program resumes

Priority Schemes

1. Software Priority

Priority determined by software instructions.

2. Hardware Priority

Priority determined by hardware circuits.

Methods:

- Daisy chain priority
- Parallel priority interrupt

Advantages:

Disadvantages:

Marking Scheme

- Definition – 2 marks
- Steps in interrupt handling – 7 marks
- Priority schemes – 4 marks
- Advantages/disadvantages – 2 marks